ESP32 Voltage Protector - Wokwi

Design and demonstration report | 10 October 2026

Design

This project demonstrates an ESP32 appliance protector in Wokwi. Two potentiometers emulate processed voltage and current readings; a green LED powered through relay contacts represents the appliance. An I2C LCD shows measurements and settings. Four buttons provide local configuration and reset.

Protection behavior

Default limits are 190-250 V and 10 A, with a 3000 ms reconnect delay. Voltage outside the allowed range opens the simulated relay. Returning to healthy readings starts a fresh delay. Over-current latches the relay open until reset, which is accepted only when both readings are safe. Zero current is allowed. Settings are validated before application and remain volatile.

Communication and implementation

The ESP32 connects to Wokwi-GUEST and a public MQTT test broker. MQTT Explorer reads JSON status and sends STATUS, SET and RESET commands. SET changes all four settings after validation. Local MENU/UP/DOWN controls use the same validation. A separate network task transfers commands through queues; the main loop owns protection state. LCD updates write only changed characters to reduce delays.

Observed results

During development the applicant reported working sensor controls, LCD, buttons, fault outputs, voltage trips, current latch/reset and remote settings changes. Supplied screenshots show MQTT telemetry at 229.9 V / 1.5 A with LOAD ON and a retained current latch after current was reduced. These are functional simulation observations; no automated ESP32 test suite or hardware timing measurements are claimed.

Limitations and next work

The project does not simulate an AC waveform or calculate RMS from physical sensor samples. Calibration, isolation, relay opening time and fast-surge protection are unvalidated. The public MQTT broker has no authentication or encryption; it is for demonstration only. Settings are not persistent. Sampling is approximately every 20 ms without a proven worst-case deadline. Additional fault/settings evidence and timing tests should be captured before submission. Physical work must use low-voltage stand-ins.

Applicant review

This is an assisted factual draft. Review the results, add only personally verified observations, and rewrite the report in your own words before submitting.